Multi-User Extension for Active STAR-RIS Assisted UAV Networks With NOMA

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This supplementary note presents a simplified $2K$-user extension of the proposed active STAR-RIS assisted UAV network with NOMA. In the main paper, the analytical framework is developed for a two-user paired NOMA scenario, where one user is located on the reflection side and the other user is located on the refraction side. To further illustrate the scalability of the proposed framework, we consider a specified multi-user case with K reflection-side users and K refraction-side users.

I. Simplified $2K$-user Signal Model

In the considered $2K$-user case, UAV transmits a superposed NOMA signal containing all users' messages, which is written as

$$X = \sum_{k=1}^{2K} \sqrt{a_k P} x_k \tag{1}$$

where x_k is the normalized unity power signal of the k-th user and a_k denotes the corresponding transmission power allocation coefficient satisfying $\sum_{k=1}^{2K} a_k = 1$.

According to the treatment adopted in **Ref.[1]**, the SIC decoding order is predefined according to the ordered power allocation coefficients, rather than dynamically optimized. The transmission power allocation coefficients are sorted as

$$a_1 < \cdots < a_s < ... < a_K < a_{K+1} < \cdots < a_t < ... < a_{2K} \tag{2}$$

where $a_1, \cdots, a_K$ correspond to the reflection-side users and $a_{K+1}, \cdots, a_{2K}$ correspond to the refraction-side users.

Similar to **Ref.[2]**, we employ geometric series power distribution method to define a_k, which is written as

$$a_k = \frac{r^{k-1}}{\sum_{t=0}^{2K-1} r^t} \tag{3}$$

where r denotes the common ratio of the geometric series and is used to reflect the non-uniformity of power allocation among users.

Under this setting, the refraction-side users are assigned higher power coefficients, while the reflection-side users are assigned lower power coefficients. The SIC decoding order is therefore determined by the ordered power allocation coefficients. In particular, a user with a lower power coefficient needs to decode the messages of higher-power users before decoding its own message. These higher-power signals may include not only the signals of users located on the same side, but also those of users located on the other side.

II. SINR Expressions for two-side users

The cascaded channel coefficients for a reflection-side user $\mathcal{U}_s$ and a refraction-side user $\mathcal{U}_t$ can be respectively written as

$$\chi_s = \mathbf{h}\mathbf{\Theta}_s\mathbf{g}_s, \quad \chi_t = \mathbf{h}\mathbf{\Theta}_t\mathbf{g}_t \tag{4}$$

Therefore, the SINR at a reflection-side user $\mathcal{U}_s$ for decoding the message of $\mathcal{U}_{i_2}$ $(s \leq i_2)$ and its own message are respectively expressed as

$$\gamma_{s\to i_2} = \frac{\rho^2 a_{i_2} P \Phi_s^2 |\chi_s|^2}{\rho^2 P \Phi_s^2 |\chi_s|^2 (S_{k<i_2} + \beta_{RI} S_{k>i_2}) + \Delta I_s} \tag{5}$$

$$\gamma_{s\to s} = \frac{\rho^2 a_s P \Phi_s^2 |\chi_s|^2}{\rho^2 P \Phi_s^2 |\chi_s|^2 (S_{k<s} + \beta_{RI} S_{k>s}) + \Delta I_s + \beta_{RI} P \sigma_e^2} \tag{6}$$

where $\chi_s = \mathbf{h}\mathbf{\Theta}_s\mathbf{g}_s$, $\Delta I_s = \rho^2\sigma_1^2 d_{ss}^{-o_s}|G_s|^2 + P\sigma_e^2 + \sigma_0^2$, $G_s = \mathbf{\Theta}_s\mathbf{g}_s$, $S_{k<i_2} = \sum_{k=1}^{i_2-1} a_k$, $S_{k>i_2} = \sum_{k=i_2+1}^{2K} a_k$, $S_{k<s} = \sum_{k=1}^{s-1} a_k$ and $S_{k>s} = \sum_{k=s+1}^{2K} a_k$. Notably, o_s is the path loss exponent of the distance d_{ss} between STAR-RIS and $\mathcal{U}_s$.

Similarly, the SINR at a refraction-side user $\mathcal{U}_t$ for decoding the message of $\mathcal{U}_{i_1}$ $(t \leq i_1)$ and its own message are respectively given by

$$\gamma_{t\to i_1} = \frac{\rho^2 a_{i_1} P \Phi_t^2 |\chi_t|^2}{\rho^2 P \Phi_t^2 |\chi_t|^2 (S_{k<i_1} + \beta_{RI} S_{k>i_1}) + \Delta I_t} \tag{7}$$

$$\gamma_{t\to t} = \frac{\rho^2 a_t P \Phi_t^2 |\chi_t|^2}{\rho^2 P \Phi_t^2 |\chi_t|^2 (S_{k<t} + \beta_{RI} S_{k>t}) + \Delta I_t + \beta_{RI} P \sigma_e^2} \tag{8}$$

where $\chi_t = \mathbf{h}\mathbf{\Theta}_t\mathbf{g}_t$, $\Delta I_t = \rho^2\sigma_1^2 d_{st}^{-o_t}|G_t|^2 + P\sigma_e^2 + \sigma_0^2$, $G_t = \mathbf{\Theta}_t\mathbf{g}_t$, $S_{k<i_1} = \sum_{k=1}^{i_1-1} a_k$, $S_{k>i_1} = \sum_{k=i_1+1}^{2K} a_k$, $S_{k<t} = \sum_{k=1}^{t-1} a_k$ and $S_{k>t} = \sum_{k=t+1}^{2K} a_k$. Notably, o_t is the path loss exponent of the distance d_{st} between STAR-RIS and $\mathcal{U}_t$.

III. Performance Evaluation

To calculate the analytical expressions of downlink metrics for active STAR-RIS assisted UAV network with multiple users, the same derivation method provided in the main paper can be followed after replacing the two-user SINR expressions with the above user-specific SINR expressions.

Using the same parameters setting of the second experiment group ($N = 600$, Section III.B) and setting $r = 1.5$, **fig. 1 exhibits the influence of number of users K_u ($K_u = 2K$) on the downlink performance of active STAR-RIS assisted UAV networks with multiple users**. It should be noted that the case of $K_u = 2$ can be regarded as the special two-user case of the considered multi-user model. However,

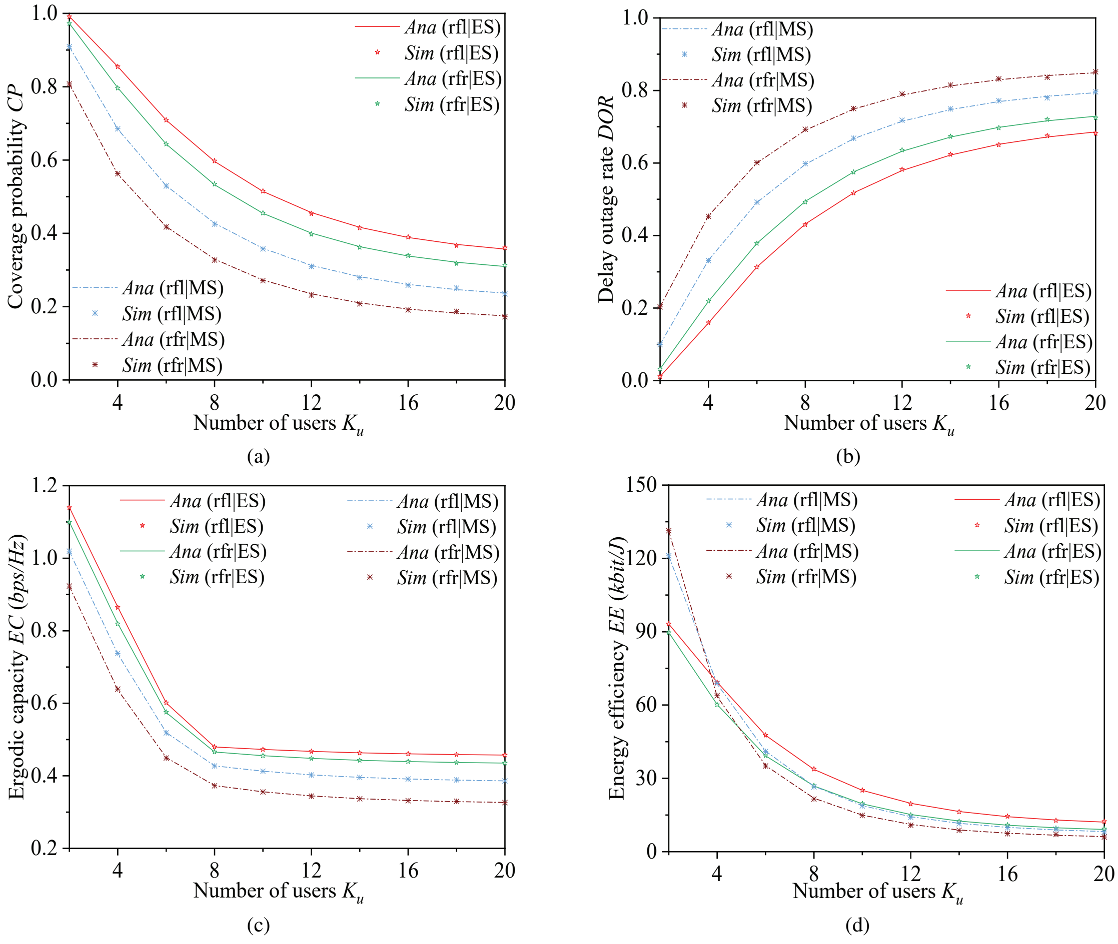

Fig. 1. The impact of number of users K_u

due to the different power allocation setting, the corresponding performance would not be identical to that of the proposed model in the main paper.

It can be observed that increasing the number of users K_u degrades the downlink performance of active STAR-RIS assisted UAV network with multiple users. Specifically, the coverage probability CP, ergodic capacity EC and energy efficiency EE decrease while the delay outage rate DOR increases. This is because when more users are simultaneously served, the inter-user interference becomes more severe and the SIC process becomes more complicated. As a result, the received SINR of each user is reduced.

Similarly, the downlink performance in the reflection region is better than that in the refraction region and executing ES protocol yields superior performance compared to executing MS protocol, which is consistent with the analysis in the main paper. Moreover, it can be seen that the performance curves gradually become flatter as the number of users K_u increases. The reason is that adding more users causes strong interference at first, leading to rapid performance degradation. When K_u becomes large, the system is already heavily affected by inter-user interference, so the additional impact of more users is smaller. Consequently, the curves become flatter. This effect is especially clear for EC, since the operator $\log_2(1+\gamma)$ varies more slowly when the SINR is in low regime.

Through comparing the simulation results with numerical analysis results, it can be drawn that the simplified $2K$-user extension is effective for depicting the downlink performance of active STAR-RIS assisted UAV networks with multiple users under the considered settings.

Ref.[1] and **Ref.[2]** are listed as follows:

Ref.[1]: X. Li, X. Gao, L. Yang, H. Liu, J. Wang and K. M. Rabie, "Performance Analysis of STAR-RIS-CR-NOMA-Based Consumer IoT Networks for Resilient Industry 5.0," *IEEE Trans. Consum. Electron.*, vol. 70, no. 1, pp. 1380-1391, Feb. 2024.

Ref.[2]: C. Fan, X. Zhou, T. Zhang, W. Yi and Y. Liu, "Cache-Enabled UAV Emergency Communication Networks: Performance Analysis With Stochastic Geometry," *IEEE Trans. Veh. Technol.*, vol. 72, no. 7, pp. 9308-9321, Jul. 2023.